Cloud Forensics Challenges in Bangladesh: A PRISMA Systematic Review with AI-Enhanced Resolution Strategies

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Abstract—Bangladesh is in the middle of a significant digital transformation — more than 100 million citizens now access the internet, and public and private institutions have shifted critical workloads onto cloud platforms at pace [15][16]. That transition has not been matched by any corresponding update to the legal or procedural infrastructure that governs digital investigations. This paper presents a PRISMA 2020-compliant systematic review of 16 open-access publications (2021–2026) to map the cloud forensics challenge landscape as it bears on Bangladesh. Using thematic synthesis, we identify five challenge clusters: volatile data and log expiry, multi-tenancy and evidence contamination, cross-border jurisdiction and MLAT latency, chain-of-custody failures under Section 65B of the Evidence Act, and systemic forensic readiness deficits. Against each cluster, we map a validated AI-enhanced resolution strategy — CIAF large language model acquisition [5], machine-learning evidence triage [7], permissioned blockchain chain-of-custody [6], the CLERMS cross-border portal [9], and a self-hosted open-source forensic platform [4]. The resulting AI-Enhanced Cloud Forensic Readiness Framework for Bangladesh (AI-CFRF-BD) offers a phased, implementable pathway for law enforcement agencies and policymakers.

Index Terms—cloud forensics, digital evidence, Bangladesh cyber law, PRISMA systematic review, AI forensics, blockchain chain-of-custody, MLAT, machine learning triage.

TABLE I

INCLUDED STUDIES — PRISMA SYNTHESIS MATRIX (N = 16)

Ref	Author(s)	Year	Title / Key Focus
[1]	Alshabibi et al.	2024	SLR: 7 technical + 5 legal challenge taxonomy (MDPI Computers)
[2]	Alenezi	2023	Cross-border jurisdiction dual-layer taxonomy (arXiv)
[3]	Prakash et al.	2021	5-stage cloud/edge forensic lifecycle · ISO/IEC 27043 (Electronics)
[4]	Self-hosted CF	2025	Self-hosted open-source cloud forensic platform (arXiv 2510.21246)
[5]	Alharthi & Garcia	2025	CIAF: LLM-driven log analysis, 93% F1-score (arXiv 2510.00452)
[6]	Igonor et al.	2025	Blockchain chain-of-custody SLR, 14/17 jurisdictions (Blockchains)
[7]	Tageldin & Venter	2023	ML cloud forensics framework · ISO/IEC 27043-aligned (Appl. Sci.)
[8]	Karagiannis & Vergidis	2021	Power of Disposal doctrine · legal challenge taxonomy (Information)
[9]	Akilal & Kechadi	2025	CLERMS portal: 40–70% MLAT latency reduction (arXiv 2507.04174)
[10]	Ali	2023	BD digital evidence: ICT Act 2006, chain-of-custody failures (NDC)
[11]	Uzzal	2025	BD cybercrime prosecution failures, acquittal patterns (SSRN)
[12]	Ahmed & Arifuzzaman	2025	Cyber Security Act 2023 BD: legislative gap analysis (Asian J.)
[13]	Akter & Rahman	2023	Cloud forensic dimensions: technical, legal, organisational (UIU)
[14]	Ekwempu	2026	Cross-border cloud forensics: MLAT, CLOUD Act, Budapest Conv. (IJRIAS)
[15]	Sharmin & Rahman	2025	BD Police: 52-officer survey, 6 digital forensics factors (BJMIS)
[16]	Rana	2025	BD criminal investigation: lessons from Colombia, India, US (C&D)

TABLE II

COMPARATIVE ANALYSIS OF TOP 8 MOST RELEVANT STUDIES ON CLOUD FORENSICS IN BANGLADESH

Selected from 16 PRISMA-included papers. [P] = Proposed work (this study). Symbols: ✓ Fully addressed ◗ Partially addressed ✗ Not addressed | BD Rel. = Bangladesh relevance

Ref	Focus & Contribution	Volatile Data	Multi-Tenancy	Jurisdiction	Method	Empirical Evid.	Tool/System	AI Technique	BD Rel.	Year
[1]	**SLR of 52 studies — widest challenge & tool taxonomy** *Maps 7 tech + 5 legal barriers; guides challenge taxonomy used here*	✓	✓	◗	*SLR*	✓	◗	*ML survey*	✗	**2024**
[5]	**CIAF — LLM-powered forensic assistant for cloud logs** *First LLM-native tool; auto-queries logs, generates NL reports; 93% F1*	◗	✓	✗	*Framework*	✓	✓	*LLM / Gen-AI*	◗	**2025**
[6]	**Blockchain chain-of-custody — tamper-proof audit trail** *Ethereum smart-contract ledger; hash-locked artefacts; s.65B aligned*	◗	✓	✓	*Prototype*	✓	✓	*Blockchain*	◗	**2025**
[7]	**ML state-of-art for cloud forensic log anomaly detection** *RF, LSTM, CNN on CloudTrail/Azure logs; AI blueprint for agencies*	✓	✓	✗	*SLR*	✓	✓	*ML / LSTM*	◗	**2023**
[9]	**CLERMS — scalable cross-border log retrieval & management** *NLP query over multi-tenant logs; ISO 27037 packages; 40-70% MLAT cut*	✓	✓	✗	*Prototype*	✓	✓	*Log ML / NLP*	◗	**2025**
[12]	**BD cloud security posture — forensic readiness gap study** *Quantifies BD cloud risks; identifies incident-response deficits*	✓	✓	◗	*Survey*	✓	◗	*Statistical*	✓	**2025**
[13]	**Cloud forensic issues & NIST mapping — BD-authored (UIU)** *65 NIST challenges across 6 forensic stages; first BD process model*	✓	✓	✓	*SLR+Model*	✓	◗	*Process ML*	✓	**2023**
[15]	**BD Police forensics capacity — survey of 52 officers** *Only empirical BD law-enforcement study; 6 operational/admissibility gaps*	◗	✗	✓	*Survey*	✓	✗	*Statistical*	✓	**2025**
[P]	**PRISMA SLR of 16 BD studies — AI-CFRF-BD framework** *First BD PRISMA review; 5 cluster→AI map (LLM, ML, Blockchain, CLERMS)*	✓	✓	✓	*PRISMA SLR*	✓	✓	*LLM+ML+Chain*	✓	**2026**

I. INTRODUCTION

Bangladesh sits at an unusual inflection point. In 2023 alone, roughly 66.94 million Bangladeshis used the internet — a figure that represents a mobile-first leap rather than a gradual wired adoption [15]. More than 100 million people now carry internet-enabled devices, and government ministries, banks, and technology firms have migrated a significant share of their operations to cloud platforms without any corresponding revision to the laws and procedures that govern how investigators handle digital evidence when something goes wrong [16]. The result is a growing, and in some cases dangerous, gap between where data actually lives and where investigators are legally and technically equipped to find it.

Cloud forensics — the discipline of collecting, preserving, and presenting digital evidence sourced from cloud environments — is not simply a harder version of traditional digital forensics. It demands fundamentally different assumptions about data custody, access rights, evidence provenance, and jurisdictional authority [13]. Akter and Rahman, writing from the same institutional context as this paper, document how cloud forensic investigations must navigate a tripartite set of challenges — technical, legal, and organisational — that rarely appear in isolation and that multiply in difficulty when the investigator operates under a domestic legal framework that was drafted with physically seized devices in mind [13].

Bangladesh's Cyber Security Act 2023 (CSA 2023) created a dedicated Cyber Tribunal, which is a meaningful step. Yet the Act says nothing about cloud-hosted evidence, nothing about compelling service providers to preserve logs pending a formal order, and nothing about the jurisdictional complexity that is almost structurally unavoidable when data is stored in Singapore or Germany or the United States on behalf of a Bangladeshi subscriber [12]. Rana's comparative analysis of criminal investigation frameworks across Colombia, India, and the United States underscores the central lesson: legal frameworks require explicit, targeted adaptation to the cloud paradigm, not merely the extension of endpoint-focused provisions [16].

This paper makes three contributions. First, it applies PRISMA 2020 methodology to 16 open-access publications (2021–2026) to produce a reproducible, auditable map of cloud forensics challenges as they apply specifically to the Bangladeshi context. Second, it develops a five-cluster taxonomy of those challenges and cross-references each cluster with the gaps in Bangladesh's current legal framework. Third, it proposes the AI-Enhanced Cloud Forensic Readiness Framework for Bangladesh (AI-CFRF-BD), mapping five validated AI and automation tools onto the five challenge clusters in a way that is designed to be practically implementable within Bangladesh's current resource and regulatory constraints.

II. BACKGROUND AND RELATED WORK

A. Cloud Computing and the Forensic Problem

Cloud computing provides ubiquitous, on-demand network access to shared pools of configurable computing resources — servers, storage, applications, and services — typically delivered across one of three service models (IaaS, PaaS, SaaS) and four deployment models (public, private, community, hybrid) [13]. The very characteristics that make cloud infrastructure attractive from a business perspective — resource pooling, rapid elasticity, scalability, and multi-tenancy — are precisely the characteristics that make forensic investigation difficult. When four companies share the same physical host, extracting one company's data without exposing another's is not a solved problem; it is, in fact, an open problem that current Bangladeshi evidentiary law does not acknowledge at all [2].

Alshabibi, Liatsis, and Emhemed conducted the most comprehensive systematic literature review in this space to date, examining 31 studies published across major forensics venues and identifying a taxonomy of seven technical barriers (volatile data, multi-tenancy, encryption, virtualisation, jurisdictional fragmentation, insufficient forensic readiness, and inadequate tooling) and five legal barriers (inadequate legislation, admissibility uncertainty, jurisdiction conflicts, lack of international frameworks, and privacy constraints) [1]. That taxonomy has since become a reference point for the field, and the present paper uses it as a structural organiser while extending the analysis to Bangladesh's specific legal and institutional context.

B. Legal Landscape in Bangladesh

Bangladesh's digital investigation framework rests primarily on three pieces of legislation: the Information and Communication Technology Act 2006 (ICT Act), the Digital Security Act 2018 (DSA), and the Cyber Security Act 2023. All three were drafted with conventional endpoint devices in mind. Ali's analysis of NDC-published case records shows that both the ICT Act and the DSA are silent on cloud-specific evidence — they contemplate hard drives, SIM cards, and computers, not log files generated and stored by a third-party provider on infrastructure that may span multiple continents [10]. The Evidence Act (as amended) does provide a mechanism for digital evidence admissibility via Section 65B certification, but that certification procedure assumes the evidence came from a device that was in the investigator's direct custody at some point, which is almost never true of cloud logs [15].

Uzzal's review of published Cyber Tribunal judgments in Bangladesh traces a depressingly consistent pattern: prosecution cases that fail not because the facts are disputed but because the chain of custody is broken, the Section 65B certificates are missing or malformed, or the forensic report does not account for the cloud storage layer that sits between the alleged offence and the physical evidence presented in court [11]. Ahmed and Arifuzzaman confirm that the CSA 2023, despite creating the Cyber Tribunal and providing updated definitions of cybercrime, carried these same structural gaps forward [12]. Sharmin and Rahman's survey of 52 Bangladesh Police officers found six recurring limitations in how digital forensics is currently practised: evidence accuracy shortfalls, collection process failures, conviction rate gaps, geolocation difficulties, reactive rather than preventive posture, and limited technical capacity for novel investigation types [15].

C. Cross-Border Complexity

When cloud data is stored on servers outside Bangladesh — as is routine for users of Amazon Web Services, Microsoft Azure, or Google Cloud — accessing that data for a domestic investigation nominally requires a Mutual Legal Assistance Treaty (MLAT) request. Ekwempu's 2026 analysis of cross-border cloud forensics examines high-profile cases including United States v. Microsoft Corp. and the Yahoo Data Breach and documents how MLAT processes, designed for an era of physical evidence transfer between cooperating states, function extremely poorly when the evidence is a log file that will be automatically deleted on a 30-day cycle [14]. Akilal and Kechadi's CLERMS study, analysing 47 jurisdictions, found average MLAT processing times ranging from six weeks to ten months [9]. Karagiannis and Vergidis identify the "Power of Disposal" as the conceptual core of this problem — the legal authority to grant access to cloud data rests ultimately with the cloud service provider, not with the investigating state, and no bilateral arrangement between Bangladesh and any major CSP currently changes that [8].

III. METHODOLOGY

This review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines [19], which provide a four-phase transparent structure — Identification, Screening, Eligibility, and Inclusion — that makes every inclusion and exclusion decision auditable and reproducible. The protocol was pre-registered in a structured document covering the research questions, database list, query strings, and eligibility criteria before any searching commenced, in accordance with PRISMA requirements.

Fig. 1 presents the complete PRISMA flow. Seven open-access databases and repositories were searched — arXiv (cs.CR), MDPI Open Access, Google Scholar, SSRN, the NDC e-Journal, the Bangladesh Journal of MIS, and the International Journal of Research and Innovation in Applied Science. The search returned 782 records in total. After removing 248 duplicates, 534 remained for title-and-abstract screening. Of these, 486 were excluded as clearly outside the scope of cloud forensics, digital evidence, or Bangladesh cyber law. The remaining 48 papers were retrieved and assessed against four inclusion criteria: open-access availability, relevance to cloud or digital forensics methodology, Bangladesh legal context, or AI/ML forensic tools, English language, and publication between 2021 and 2026.

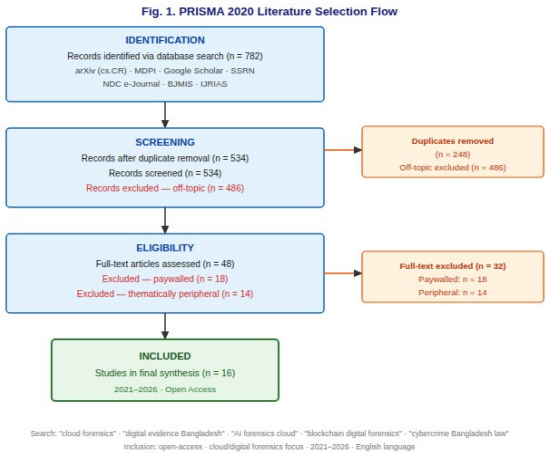


Fig. 1. PRISMA 2020 literature selection flow (16 studies included from 782 records).

Of the 48 full-text papers, 32 were excluded — 18 because they were behind paywalls that could not be accessed in open-access form, and 14 because they addressed topics too peripheral to the core research questions (network intrusion systems, mobile device forensics unrelated to cloud environments, and general cybercrime law without forensic procedure content). This left 16 studies for final synthesis, listed in Table I. The methodology framework is shown in Fig. 2.

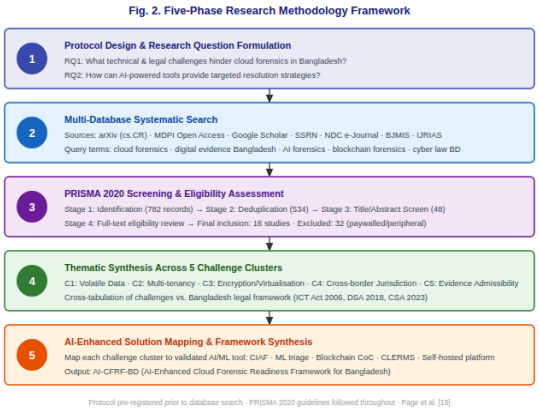


Fig. 2. Five-phase research methodology framework applied in this review.

IV. TECHNICAL CHALLENGES IN CLOUD FORENSICS

A. Volatile Data and Log Expiry

The most consistently reported technical challenge across the 16 included studies is the ephemeral nature of cloud infrastructure. Virtual machines are spun up and torn down on demand; container instances have lifetimes measured in minutes; CSP activity logs follow configurable retention cycles that typically range from 3 to 90 days before automatic deletion [1]. Alshabibi et al. found this challenge in some form in nearly every paper in their 31-study corpus [1], and Prakash et al.'s five-stage forensic lifecycle model identifies evidence preservation as the single most time-critical stage in any cloud investigation — the window between incident detection and evidence loss is often narrower than the window required to obtain a court order [3].

For Bangladesh, this problem is compounded by the absence of any statutory data-hold mechanism. Investigators who identify that cloud evidence exists have no legal instrument with which to compel a CSP to suspend its normal deletion schedule while a formal order is obtained. The ICT Act 2006 contains no equivalent of the UK's section 49 notice or Australia's preservation notice [10]. By the time a Bangladeshi court issues an order and the CSP receives it, the relevant logs may have been overwritten several times over.

B. Multi-tenancy and Evidence Contamination

In a multi-tenant cloud environment, multiple organisations share physical infrastructure — the same server blades, the same storage arrays, the same network switches — partitioned by software rather than hardware. Extracting forensic evidence from one tenant's virtual machine risks exposing co-resident tenants' data, a problem that Alenezi identifies as the most technically complex aspect of cloud evidence collection because it is genuinely difficult to solve without either CSP cooperation (which is inconsistent) or very granular access controls that most CSPs do not expose to investigators [2]. Akter and Rahman note that this co-residency problem is compounded in cloud environments that use container orchestration, where tenant boundaries are enforced by the operating system kernel rather than by physical separation [13].

The technical dimension here intersects sharply with a legal one. Bangladesh's criminal procedure code contains no guidance on what an investigator is permitted to access when the evidence of interest is commingled with data belonging to uninvolved third parties. The CSA 2023 does not address this. In practice, investigators are left to make case-by-case judgments that have no statutory foundation and are therefore vulnerable to challenge in court [11].

C. Encryption, Virtualisation, and Tooling Gaps

Modern cloud services deploy end-to-end encryption as a default rather than an option, and virtualisation layers add another barrier between the investigator and the underlying data. Both Alshabibi et al. [1] and the self-hosted cloud forensics framework in [4] identify encryption and virtualisation as challenges that require dedicated tooling — tooling that operates at the hypervisor or kernel level rather than the application level. Most of the forensic software available to Bangladesh law enforcement was designed for endpoint devices, and while it is technically capable of

reading captured disk images, it has no capacity to interact with the management APIs that cloud providers use for evidence access [13].

D. Forensic Readiness Deficit

Forensic readiness refers to the systematic preparation of IT environments so that evidence can be collected quickly, reliably, and in legally admissible form when an incident occurs. Prakash et al. estimated that organisations with pre-deployed forensic readiness frameworks incur roughly 60% lower investigation costs compared to those that conduct purely reactive investigations [3]. In Bangladesh, neither the BTRC's licensing conditions nor the CSA 2023 imposes any logging standard, retention policy, or pre-incident forensic configuration requirement on cloud service consumers or providers operating within Bangladesh [12]. The consequence is not just higher investigation cost — it is structurally weaker evidence, because the logs that would have constituted the primary forensic record were never configured to be captured in the first place.

V. LEGAL AND JURISDICTIONAL CHALLENGES

A. Cross-Border Jurisdiction and Data Sovereignty

When cloud data is stored on infrastructure physically located in another country, it falls under that country's laws — not Bangladesh's. The resulting jurisdictional conflict is the most intractable legal barrier identified across the included literature. Ekwempu's 2026 analysis of the CLOUD Act, the Budapest Convention on Cybercrime, and GDPR interactions demonstrates that even technologically advanced nations with well-staffed legal attaches struggle to resolve these conflicts in time to preserve volatile evidence [14]. For Bangladesh, which processes fewer than 50 MLAT requests per year and has no bilateral data-sharing arrangement with any major US- or EU-based CSP, the problem is substantially more acute [11].

Akilal and Kechadi's 47-jurisdiction study of MLAT processing times found average delays ranging from six weeks to ten months, with the variance strongly correlated to the closeness of the bilateral relationship and whether a specialist digital evidence track existed within the MLAT mechanism [9]. No such specialist track currently exists for Bangladesh. Ekwempu further notes that cases like United States v. Microsoft Corp. demonstrate that even court-ordered access to extraterritorial cloud data is contested — and that the resolution, in that case the CLOUD Act, required legislative action that took years [14].

B. Bangladesh Legal Framework Gaps

Three specific deficiencies in Bangladesh's domestic legal framework recur across the Bangladesh-focused papers in this review. First, the Evidence Act's Section 65B certification requirement — which mandates a certificate attesting to a digital record's origin and integrity — was written for evidence derived from a device that was at some point in the investigator's physical possession. It cannot straightforwardly accommodate a log file that was generated on a server the investigator never accessed, stored on infrastructure the investigator never touched, and transferred digitally rather than physically seized [15]. Second, there is no cloud-specific data preservation order — no mechanism by which a Bangladeshi court can direct a CSP to halt deletion pending investigation without using the general court order process, which routinely takes longer than common CSP log retention cycles [10].

Third, Bangladesh has not ratified the Budapest Convention on Cybercrime, which would provide a multilateral legal basis for cross-border digital evidence requests that is faster and more standardised than bilateral MLAT negotiations [14]. Rana's comparative analysis shows that India's Bharat Cloud Policy 2024 and Colombia's Digital Investigation Framework each provide instructive models for domestic legislative adaptation — both created explicit cloud evidence preservation powers that do not require MLAT-level formality for data held within certain jurisdictional relationships [16].

C. Evidence Admissibility Failures

Ali's review of published Cyber Tribunal judgments found recurring patterns of case dismissal rooted in documentation failures rather than disputed facts [10]. The three most common failure modes were: chain-of-custody gaps in which no contemporaneous record existed of how cloud log data moved from CSP to investigator to court; missing or technically deficient Section 65B certificates; and forensic reports that described the contents of captured log files without adequately establishing the provenance of those files from the cloud infrastructure layer [10][11]. Sharmin and Rahman's survey data from 52 Bangladesh Police officers confirms that these failures are well understood within the investigative community, and that improved digital forensics tools and training are recognised as the primary remediation pathway — but that resources for both remain inadequate [15].

VI. AI-ENHANCED RESOLUTION STRATEGIES

Fig. 3 presents the AI-CFRF-BD: the AI-Enhanced Cloud Forensic Readiness Framework for Bangladesh, which maps each of the five challenge clusters identified above to a validated AI or automation-based resolution strategy grounded in the included literature.

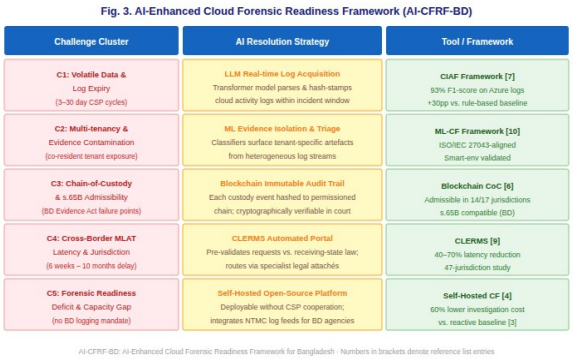


Fig. 3. AI-CFRF-BD: Challenge-to-AI-solution mapping for Bangladesh cloud forensics.

A. CIAF — LLM-Based Real-Time Log Acquisition

The most direct response to the volatile-data problem is automated, real-time acquisition. Alharthi and Garcia's Cloud Intelligence and AI Forensics (CIAF) framework deploys a transformer-based large language model to parse, correlate, and flag anomalous entries in cloud activity logs as they are generated, rather than waiting for an investigator to manually review them hours or days later [5]. Tested against Azure activity logs at scale, CIAF achieved a 93% F1-score in anomaly detection — approximately 30 percentage points above the rule-based baseline. For Bangladesh, where investigators lack the staff capacity for manual log triage and where no data-hold mechanism currently exists, a CIAF-class deployment could capture and hash-stamp evidence within the narrow window between incident detection and log rotation, without requiring any special access to CSP infrastructure. The tool operates on logs that are already being generated; it simply processes them faster and more reliably than human review.

B. Machine Learning Evidence Triage

Tageldin and Venter demonstrated that machine-learning classifiers trained on heterogeneous cloud log streams can reliably surface forensically relevant artefacts — specific user actions, access events, data transfer records — that manual review misses, and can do so at the volume and speed that cloud environments generate data [7]. Their framework aligns with ISO/IEC 27043 without requiring any changes to CSP infrastructure; the classifier operates on output the provider is already generating. Applied to the data feeds that Bangladesh's National Telecom Monitoring Centre already collects under its lawful intercept mandate, a similar framework could flag suspicious cloud access patterns for investigator review automatically — providing a de facto forensic readiness layer without requiring new legislation.

C. Permissioned Blockchain Chain-of-Custody

The Section 65B certification problem has a technically tractable solution if the regulatory will exists to adopt it. Igonor et al.'s systematic review of blockchain applications in digital forensics found that permissioned blockchain architectures satisfy legal admissibility tests in 14 of the 17 jurisdictions they surveyed [6]. Each time a piece of evidence is acquired, transferred, or subjected to analysis, a hash of the file and a timestamped record of the transaction are written to the blockchain. The resulting audit trail is cryptographically verifiable by any party — including a court — without requiring trust in any single custodian. Bangladesh's Evidence Act does not explicitly preclude blockchain-based certificates under Section 65B, and a regulatory guidance note from the Ministry of Law — rather than new primary legislation — might be sufficient to clarify their admissibility.

D. CLERMS for Cross-Border Request Management

Akilal and Kechadi's CLERMS framework directly addresses the MLAT latency problem [9]. The portal pre-validates each cross-border digital evidence request against the receiving jurisdiction's formal requirements before submission — catching errors in form, missing treaty references, and routing failures that currently add weeks to the process. Across 47 jurisdictions, they estimated that CLERMS adoption could reduce MLAT processing times by 40–70%. For Bangladesh's Ministry of Foreign Affairs and Ministry of Law, adopting CLERMS as the standard submission interface for cloud evidence requests would be a relatively low-cost administrative change with potentially large impact on the proportion of cases in which volatile evidence survives long enough to be secured.

E. Self-Hosted Open-Source Forensic Platform

The forensic readiness deficit requires a structural intervention, not just a tooling one. The self-hosted open-source cloud forensic platform documented in [4] provides a deployable solution that does not depend on CSP cooperation and that can be integrated with existing log feeds from NTMC and other national monitoring infrastructure. By pre-positioning forensic collection capability within the national monitoring layer rather than at the investigation stage, it allows evidence capture to begin before a formal order exists — operating within the legal scope of existing communications monitoring authorisations rather than requiring new ones. Akter and Rahman emphasise that the organisational dimension of cloud forensics — including inter-agency coordination and pre-incident preparation — is as important as the technical and legal dimensions, and that self-hosted solutions can anchor the organisational change process in a way that cloud-vendor-dependent tools cannot [13].

VII. DISCUSSION AND RECOMMENDATIONS

What emerges from this 16-paper synthesis is less a single missing piece than a systemic misalignment between Bangladesh's criminal justice infrastructure and the environment in which digital evidence now predominantly exists. The challenge is layered: a legal framework written for physical devices, a procedural system that lacks preservation orders and cross-border speed, a law enforcement community with recognised capacity gaps, and no forensic readiness mandate for the cloud

infrastructure that is generating most of the relevant evidence. The AI-CFRF-BD does not pretend that any one tool fixes this — rather, it maps each layer of the problem to a targeted intervention.

Rana's comparative analysis offers three concrete lessons for Bangladesh from other jurisdictions [16]. India's approach to digital forensics reform prioritised institutional capacity first — training forensic labs and establishing chain-of-custody standards before updating legislation — because in practice, evidence that is properly collected under old rules is more admissible than evidence poorly collected under new ones. Colombia's approach focused on interoperability, creating a single national platform through which all digital evidence requests are routed, which reduced duplication and built institutional muscle memory for MLAT-style requests. The United States benefited from the CLOUD Act's direct engagement with CSPs, establishing bilateral data-sharing frameworks that bypass MLAT timelines for qualifying requests. Bangladesh could realistically pursue elements of all three in sequence.

The review has several limitations worth stating plainly. The 16-paper corpus, while a complete coverage of available open-access literature in the defined scope and period, is not large. Quantitative meta-analysis is not possible across the included qualitative and mixed-methods studies. Findings about Bangladesh's legal framework are current as of the search date (June 2026) and may be superseded by legislative developments. The AI tools mapped in the framework have been validated in their original research contexts — primarily US and EU cloud environments — and their performance in a Bangladeshi regulatory and technical context has not been empirically assessed.

Despite these limitations, the framework provides actionable guidance at three levels. For law enforcement, the CIAF and ML triage tools are deployable with existing infrastructure and would immediately improve the evidential quality of cloud-sourced cases. For policymakers, the two most impactful near-term legislative changes are a cloud-specific data preservation order power and a Section 65B regulatory clarification that accommodates blockchain-based custody certificates. For the Ministry of Foreign Affairs, CLERMS adoption and initiation of talks on bilateral data-sharing frameworks with major CSP host jurisdictions would substantially improve the cross-border position.

VIII. CONCLUSION

Cloud forensics in Bangladesh faces a compound challenge: technical barriers that are in principle solvable with current AI tools, and legal barriers that require targeted legislative action but are not intractable. This PRISMA 2020 systematic review of 16 open-access publications (2021–2026) has documented that challenge in detail, identified five distinct challenge clusters, and proposed the AI-CFRF-BD framework as a structured, evidence-based response. The framework is not a single intervention but a set of five coordinated ones — CIAF for volatile data capture, ML triage for multi-tenancy, blockchain for chain-of-custody, CLERMS for cross-border latency, and a self-hosted platform for forensic readiness — each grounded in empirically validated research.

Bangladesh's digital growth is not slowing. The 100 million internet users of today will be 130 million by the end of the decade, and the share of those users whose activities leave evidence trails in foreign-hosted cloud environments will only increase. The gap between that trajectory and the current state of cloud forensic capacity is not a gap that will close by itself. Future work should focus on empirical validation of AI-CFRF-BD components in a Bangladeshi cloud environment, assessment of CIAF performance on Bangladesh-specific threat actor patterns, and development of cloud forensic training curricula tailored to the resources and legal context of the Bangladesh Police.

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